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The oral microbiome of treated and untreated chronic HCV infection: A preliminary study

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Abstract

Objectives: Chronic hepatitis C virus (HCV) infection is a debilitating disease that is lately treated using direct-acting antivirals (DAAs). Changes in the oral microbiome were detected in other liver diseases; however, oral microbiome was never investigated in patients having chronic HCV infection, whether pre- or post-treatment.

Materials and Methods: This case–control preliminary study enrolled three equal groups: Group (I): untreated HCV patients; group (II): HCV patients who achieved viral clearance after DAA administration; and group (III): healthy controls. For each participant, a buccal swab was harvested and its 16S rRNA was sequenced.

Results: The oral microbiome of chronic HCV patients had a significantly distinct bacterial community compared to healthy controls, characterized by high diversity and abundance of certain pathogenic species. These changes resemble that of oral lichen planus patients. After treatment by DAAs, the oral microbiome shifted to a community with partial similarity to both the diseased and the healthy ones.

Conclusions: Chronic HCV is associated with dysbiotic oral microbiome having abundant pathogenic bacteria. With HCV clearance by DAAs, the oral microbiome shifts to approach the healthy composition.

KEYWORDS
16S rRNA, direct acting antiviral, liver disease, oral bacteria, oral lichen planus, oral microbiome

1 | INTRODUCTION

Chronic hepatitis C virus (HCV) infection can cause severe hepatic affectation such as liver cirrhosis and hepatocellular carcinoma (H. C. Li & Lo, 2015). Therefore, chronic HCV represents a major physical and economic burden to patients (Abdel-Razek et al., 2019).

Recently, the introduction of direct-acting antiviral (DAA) therapy has reformed HCV management. It provides promising results of chronic HCV elimination after its short-term use with less side effects when compared to the previously used interferon-based regimens (El Kassas et al., 2018; Raad et al., 2018).

Proper understanding of the pathophysiology of liver disease has led to the discovery of the presence of a distant active player in the disease process, namely gut microbiota (Konturek et al., 2018). In chronic liver diseases, dysbiosis was reported in the gut microbiome in the form of an increase in the levels of pathogenic bacteria and their endotoxins. Furthermore, the intestinal wall was proven to acquire higher permeability allowing for the dysbiotic bacteria and its endotoxins to translocate to distant organs. The endotoxins, then, induce inflammatory reactions causing cytokine storm. Together with the chronic inflammation and immune dysfunction caused primarily by liver disease, the inflammatory changes caused by the gut

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involvement worsen the systemic condition and exacerbate the liver disease (Arab et al., 2020; Fukui, 2019).

It is reported that stool samples of patients with chronic liver diseases showed high abundance of oral bacteria within the gut microbiome (Olsen & Yamazaki, 2019). Accordingly, the oral microbiota was proposed as another key player in liver diseases. The oral cavity, gut, and liver are believed to be linked together in an "oral–gut–liver" axis, where each organ affects the other two and is affected by them in cases of health and disease (Acharya et al., 2017).

However, very little is known regarding the changes in the oral microbiome in cases of chronic HCV infection, while none is revealed regarding the effect of DAAs on it. Therefore, this study was performed to investigate the changes in the oral microbiome of patients with chronic HCV infection and patients treated by DAAs, in comparison with healthy controls.

2 | MATERIALS AND METHODS

2.1 | *Study settings*

This case–control preliminary study included 30 participants: 10 treatment-naive chronic HCV patients (group I), 10 former chronic HCV-positive patients who were successfully treated with DAAs (group II), and 10 healthy controls matched by age and sex (group III). Being the first study to investigate the oral microbiome of patients having chronic HCV, the sample size was estimated based on a study that investigated patients having chronic HBV (Ling et al., 2015). Based on an effect size of the Chao1 index of 1.44, power of 80%, and 0.05 accepted alpha error, each group was calculated to include nine participants. After adding one participant per group for anticipated missing data, each group included 10 participants.

Groups I and II patients were recruited from the outpatient viral hepatitis clinic in Om-Elmasreyeen Hospital, Giza, Egypt, while healthy controls were recruited from the outpatient clinic of Faculty of Dentistry, Cairo University. All the sequencing and laboratory steps were performed at Colors Medical Laboratories, Cairo, Egypt. The recruitment and sampling procedures extended through September and October 2019, while laboratory analysis proceeded till March 2020.

The study included adults (≥18 years old) and non-smokers, while excluded individuals who: (i) administered any antibiotics or probiotic through the past month, (ii) had any other systemic disease, (iii) had any known active bacterial, fungal, or other viral disease, and (iv) had clinically apparent oral mucosal disease. Pregnant and lactating females were also excluded.

Healthy controls were recruited based on the negative results of ELISA test for anti-HCV antibodies. The test was performed during the national HCV screening campaign using the WHO finger-prick-based rapid test (SD Bioline HCV, Abbott, Abbott Park, IL, USA) (Esmat et al., 2018), while diagnosis of group I and II participants was further confirmed by PCR tests.

2.2 | *Sampling procedures*

After a participant provided an informed consent, the medical and personal history were assessed and conventional oral examination (Glick et al., 2003) was performed to ensure the participant's eligibility. When eligible, the participant was asked to gently rinse with water to dislodge any debris. Sampling was performed by swabbing the buccal mucosal tissues of the participants using an autoclave-sterilized interdental brush (Oral-B PRO-EXPERT, P&G, Ohio, U.S.) through applying 10 clockwise strokes to the buccal mucosal tissues (Fourie et al., 2016; Zaura et al., 2009).

The samples were first preserved in the stabilizing solution of the OMNIgene™ oral kit (OMR-110) (DNA genotek, Ottawa, Canada), following the previously described technique (Luo et al., 2016; Vogtmann et al., 2018). It was kept at room temperature, according to the manufacturer instructions until DNA extraction. However, this technique did not yield any satisfactory results at the first polymerase chain reaction (PCR).

Therefore, the samples were shifted to be preserved in sterile brain heart infusion broth as a non-selective growth medium (Sibley et al., 2011). The samples were transported at −20°C to the laboratory as soon as possible for incubation for 24 hours; after which, DNA extraction was performed.

2.3 | *DNA extraction*

Bacterial genomic DNA was isolated and purified using PrepMan Ultrasample preparation reagent (Life Technologies, Thermo Fisher, Carlsbad, CA, USA) according to the manufacturer instructions.

2.4 | 16S primers and Amplicon library generation

PCR was performed to amplify the 16S rRNA hypervariable regions V2-4-8 and V3-6, 7–9 using the primer sets of the Ion 16S™ Metagenomics Kit (Thermo Fisher, Waltham, MA, USA). For each sample, two reactions, one for each primer, were performed. The PCR amplification products were purified using the Agencourt AMPure XP beads (Beckman Coulter, Indianapolis, IN) and 70% ethanol, according to manufacturer instructions.

The DNA input for amplicon library preparation was analyzed using the Qubit™ dsDNA HS Assay Kit (Life Technologies, Thermo Fisher, Carlsbad, CA, USA). The pooled amplicons were then repaired and purified using the Ion Plus Fragment Library Kit (Life Technologies, Thermo Fisher, Carlsbad, CA, USA) following the manufacturer instructions.

2.5 | *Template preparation and sequencing*

The concentration of the purified amplicons was then quantified using qPCR to be ready for Ion OneTouch reactions. In these

reactions, the amplicons were amplified and enriched utilizing the Ion 520 & Ion 530 Kit (Life Technologies, Thermo Fisher, Carlsbad, CA, USA) with Ion OneTouch 2 system (Thermo Fisher, Waltham, MA, USA). The amplicons-loaded Ion Torrent chips were then sequenced using Ion S5 system sequencer (Ion Torrent, Thermo Fisher, Waltham, MA, USA).

2.6 | *Bioinformatics analysis of 16S data*

The bacterial 16S rRNA reads were processed and analyzed using the QIIME software pipeline release 1.9.1 (Caporaso et al., 2010). The UCLUST method was used to pick and cluster the OTUs against the Greengenes (V 13.8) 16S rRNA gene database at 97% sequence similarity (McDonald et al., 2012). The OTUs filtration from an OTU table was carried out by removing chimeric OTUs, discarding all the singleton bacterial OTUs observed fewer than three times and eliminating the OTUs with a total observation count <100.

2.7 | *Statistical analysis*

The within-sample diversity was determined by calculating the alpha diversity metrics (Observed OTUs, Shannon, Simpson, and Chao 1). Alpha diversity values were calculated at 49,746 reads as the smallest set of sequences. Box plots were constructed, followed by calculating the *p*-value using Monte Carlo permutations as a nonparametric two-sample *t* test to compare alpha diversity values between study groups (untreated HCV+, treated with DAAs, healthy controls).

The differential abundance testing of the oral bacterial community at the family, genus, and species levels was performed to identify OTUs that differ between study groups using the DESeq2 algorithm (Love et al., 2014). The bacterial abundance differences were expressed as log2 fold change (log2FC), and only the comparisons with FDR *p*-value <0.05 were considered. The shared and unique bacterial OTUs between study groups were represented using a Venn diagram. Non-metric Multidimensional Scaling (NMDS) was used to compare groups of samples based on the count-based distance matrix using the Bray–Curtis distances, followed by analyzing the strength and statistical significance of sample groupings using ANOSIM statistical method in R package Vegan (Oksanen, 2015).

3 | RESULTS

The study included three groups of participants: patients having chronic HCV infection (group I), former chronic HCV-positive patients who have been successfully treated with DAA (group II), and healthy controls (group III). Supporting information 1 includes a flowchart clarifying the number of participants in each study step.

3.1 | Demographic parameters

The age of the participants ranged between 21 and 58 with a mean age of 34.8 years. Cases and controls were matched by age and sex; therefore, no statistically significant difference was detected in demographic criteria of participants. (Table 1).

3.2 | Diversity analysis between and within groups

About 6.8 million filtered paired-end reads of 16SrRNA sequence, clustered into 214 OTUs, were acquired by investigating 30 buccal swab samples. Table S1 summarizes statistics for the Ion S5 next-generation sequencing runs for all samples (Supporting information 2). The samples of the untreated HCV+ group had the highest numbers of OTUs that ranged from 60 to 119, while OTUs number in the samples of the healthy controls group ranged from 37 to 85. No statistically significant differences were observed between the OTUs in the patients' samples post-treated with DAAs (ranged from 49 to 108) and the healthy controls group.

Beside the OTUs level, alpha diversity was calculated using Chao, Simpson, and Shannon indices (Figure 1). However, differences in the three indices were statistically significant only between the untreated and control groups. Also, the beta diversity between samples—assessed by NMDS analysis (Figure 2)—showed that the overall oral bacterial composition of the untreated group was relatively distinct from healthy individuals (ANOSIM (R) = 0.121: p = 0.01). Moreover, the post-treated samples had a different dissimilarity and were distributed between untreated and baseline samples.

TABLE 1 Demographic characteristics of the participants

Variables	HCV untreated (group I)	HCV-DAA (group II)	Healthy Control (group III)	*P*-value[a]
Age [mean (±SD)]	36.5 (10.4)	36.8 (7.9)	31.2 (5.9)	0.25
Sex				
Males [n (%)]	3 (30)	3 (30)	2 (20)	1
Female [n (%)]	7(70)	7(70)	8 (80)	

[a]Statistical significance was tested by one-way ANOVA test.

3.3 | Oral microbiota structure profile

The bacterial community structure was analyzed based on family, genus, and species taxonomy levels to investigate the oral microbiota profile. In total, 15 families (Figure S1), 52 genera (Figure S2), and 100 species (Figure S3) were detected (Supporting information 3). In all taxonomy levels, there were no other OTUs ≧ 1% of all sequences.

3.4 | Core analysis of oral microbiome

The analysis included compositions detected in at least 50% of the samples in each group, along with revealing the unique and shared OTUs at species taxa level of the three groups (Figure 3). The untreated HCV+ group showed an increase in the number of unique bacterial species with 17 species mainly of the *Prevotella* and *Veillonella* genera, while the control group's core microbiome

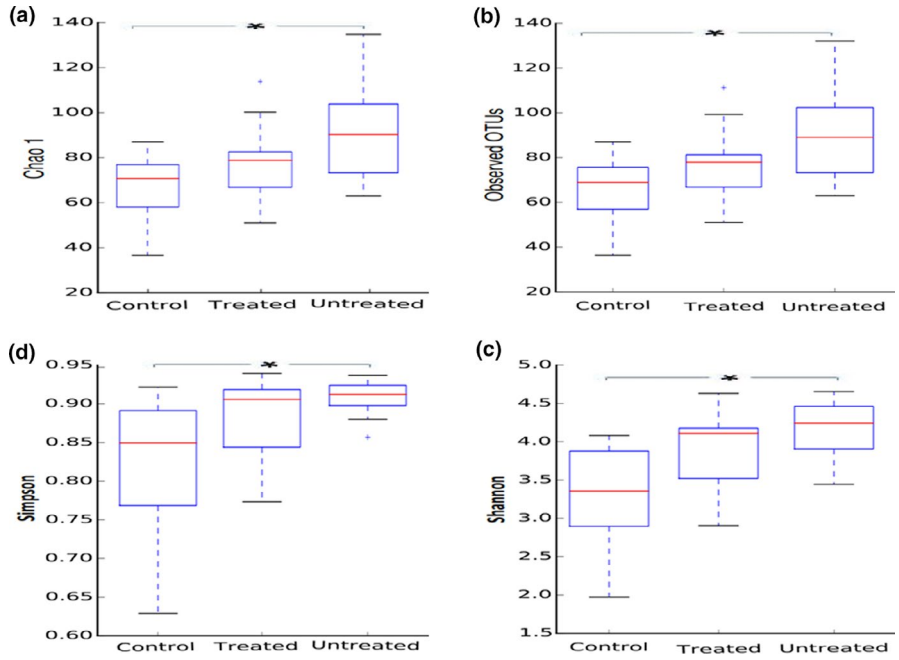


FIGURE 1 Box plots of Chao 1 (a), observed bacterial OTUs (b), Shannon (c), and Simpson (d) index values for comparison of bacterial diversity in control, treated, and untreated groups (n = 10). *p*-value was calculated by two-sample t test (*$p < 0.05$)

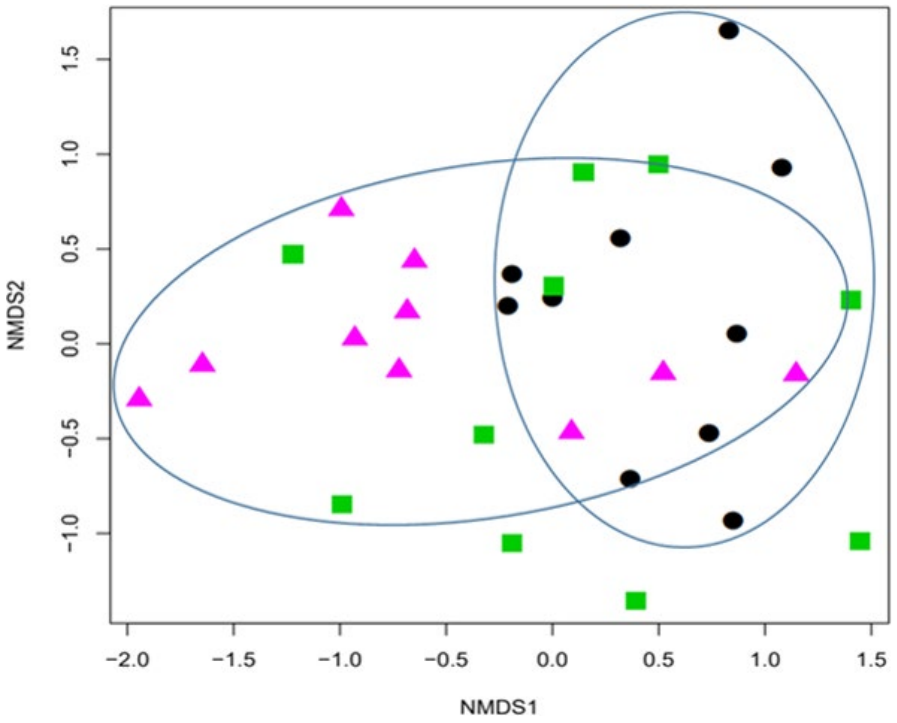


FIGURE 2 Non-metric multidimensional scaling analysis of the Bray–Curtis dissimilarity index represent the bacterial community structure based on Ion Torrent next-generation sequencing of 16S rRNA genes (Stress value: 0.13). Symbols: *untreated group*—black circle; *treated group*—green square; and *control group*—magenta triangle

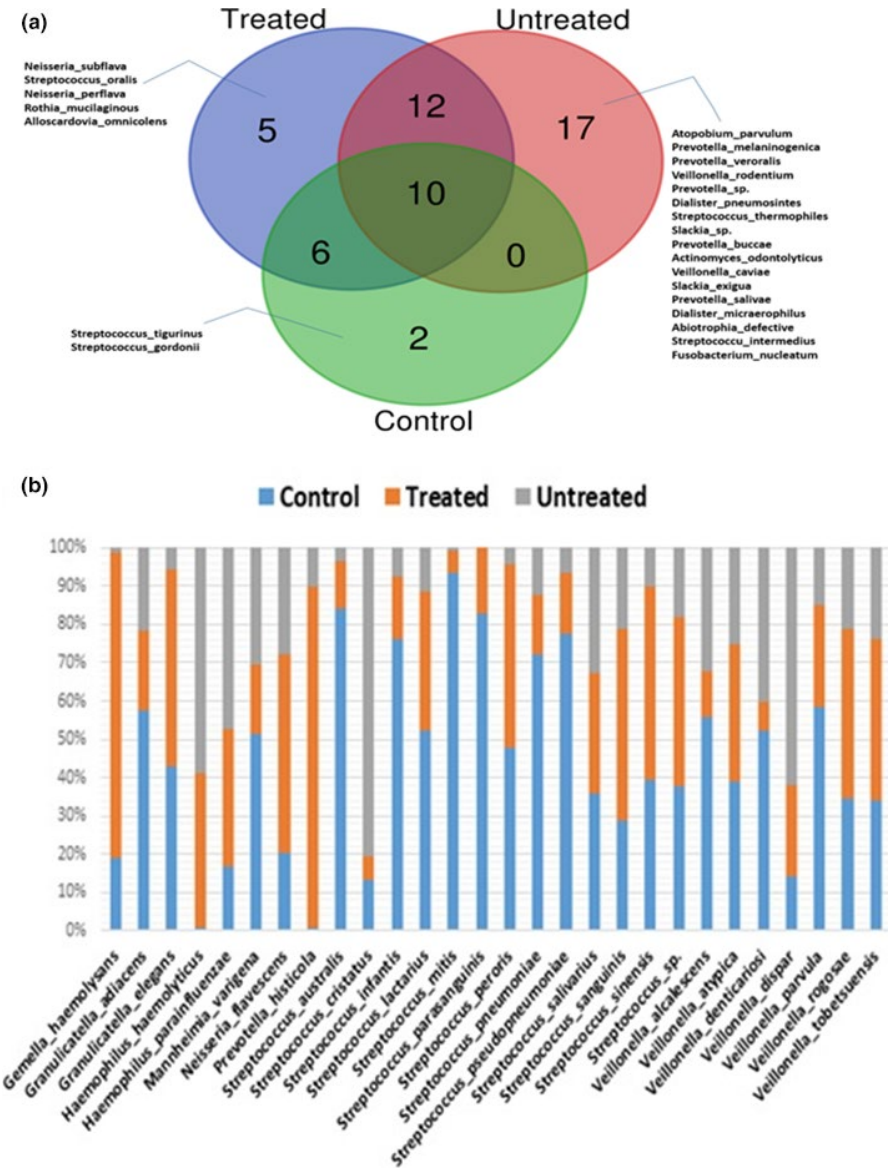


FIGURE 3 Core bacterial OTUs of analyzed samples. (a) Venn diagram representing the Core OTUs (Species level) across 50% of samples in each studied group (unique OTUs) and their intersection. (b) Abundance of the core bacterial species (based on 16S rRNA gene) that shared between studied groups

has only two unique species which belong to *Streptococcus* genus (Figure 3a). The abundance of shared bacterial species intersecting between the three studied groups was illustrated in the bar chart (Figure 3b).

3.5 | The oral microbiome differential abundance

Changes in the oral microbiome of HCV + patients and treated individuals were compared to healthy controls as a baseline. Differential abundance revealed highly statistical significant differences ($p < 0.01$) in the enriched species in treated and untreated groups, compared to healthy controls. Although the oral microbiome of the treated individuals appeared more similar to that of the healthy controls, there were some significant differences still not restored (Figure 4).

4 | DISCUSSION

Chronic HCV infection represented an active point of research for many years (Manns et al., 2017). Yet, the link of liver disease and its treatment to oral microbiome through the "oral–gut–liver" axis is not well understood (Acharya et al., 2017). This information is expected to unveil evidence about the disease pathogenesis, the implication of oral microbiome in the oral–gut–liver axis and the impact of DAAs on the oral microbiome as well (Ding & Schloss, 2014; Zhang et al., 2018).

In the present study, we investigated the oral microbiome of 10 patients having chronic HCV and 10 individuals who received treatment by DAAs, comparing it to that of 10 healthy individuals. Oral microbiome was obtained from swabbing the buccal mucosal tissues of the participants; and 16S rRNA sequencing was performed.

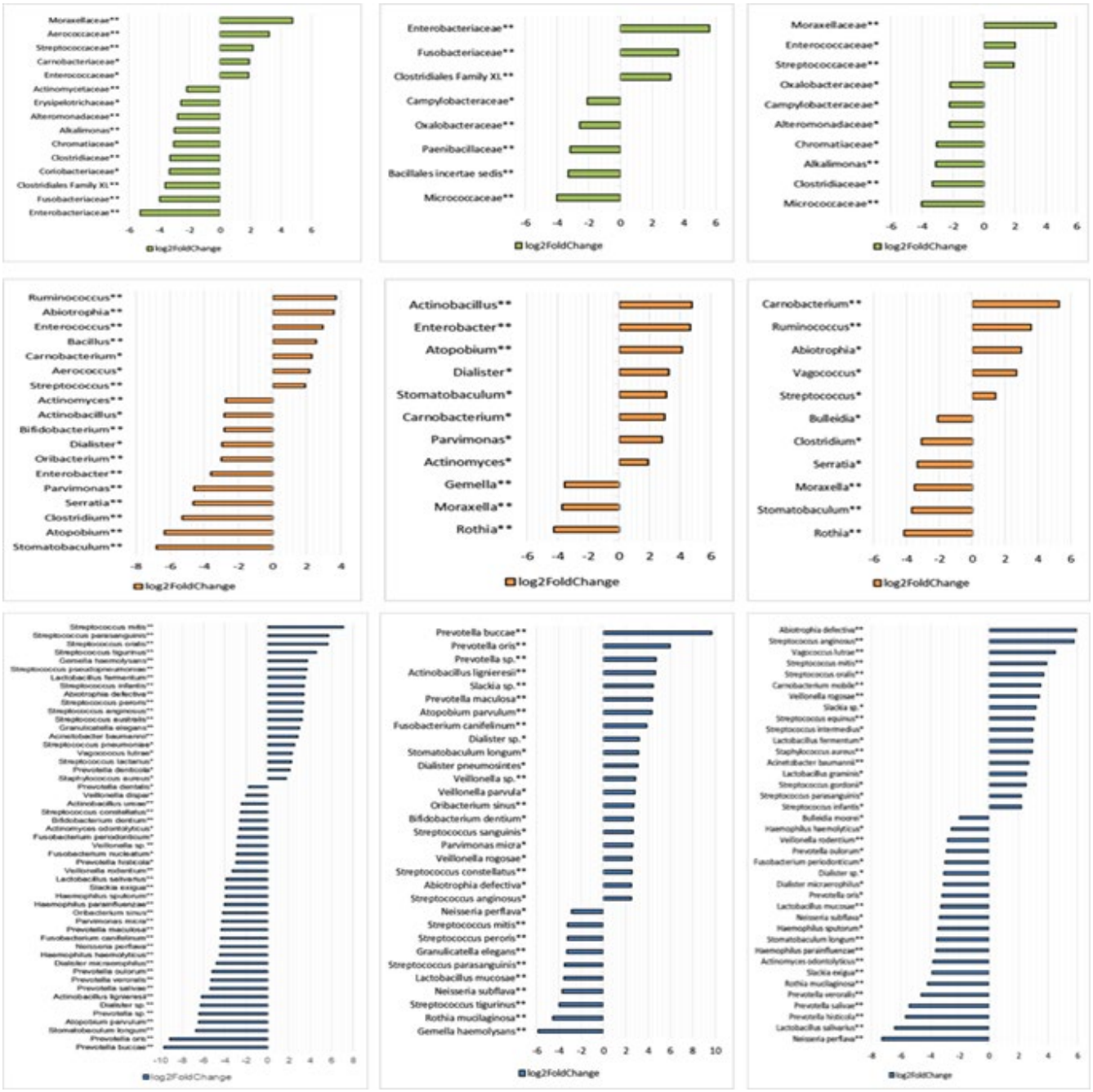


FIGURE 4 Differential oral bacterial abundance analysis at the family (upper), genus (middle), and species (lower) level between: untreated group vs. control group (left); treated group vs. untreated group (center); and treated group vs. control group (right). Negative or positive log2FC values indicate a decrease or increase in the ratio between the bacterial abundance of comparison groups. Responders with log2-fold change with the *p*-value adjusted by the Benjamini–Hochberg FDR procedure for multiple comparisons >0.05 were not presented

Sampling of the biofilm of the buccal mucosa was selected to minimize the confounded results by the periodontal health status (Zhou et al., 2013) produced by sampling of dental plaque (Haffajee & Socransky, 2001), saliva (Lira et al., 2018; Paju et al., 2009), and tongue dorsum (Arweiler & Netuschil, 2016; Göhler et al., 2018). Not only does the high rate of shedding of the buccal mucosal cells represent stresses to biofilm formation, but the direct contact with the host's innate immunity also adds to the survival challenge. Consequently, only bacteria that can confront high survival stresses successfully colonize the buccal mucosa (Samaranayake & Matsubara, 2017; Xin & Jinzhi, 2015).

Moreover, swabbing the buccal mucosa was based on the recommendation of the seventh World Workshop of Oral Medicine to sample the oral mucosal epithelium for studying the oral microbiome (Varoni et al., 2019).

The study showed that the oral microbiome profile of patients with chronic HCV had significantly higher alpha diversity indices, together with distinct bacterial community when compared to healthy

controls. On the contrary, the oral microbiome profile of individuals treated by DAAs lies in an intermediate state between that of patients having chronic HCV and healthy controls, approaching the healthy composition but not reaching it. Furthermore, a set of bacterial species were detected to increase in the microbiome of the diseased group and decline in the treated group including *S.perioris*, *S. mitis*, *S. parasanginus*, *S. tigurinus*, *G. hemolyticus*, and *Granulicatella elegans*.

The results point out to the link between the oral microbiome and the liver at states of health and disease. This clarifies that the oral microbiome is not linked to the liver condition through the periodontal pathogens only as previously hypothesized (Jasmohan S. Bajaj et al., 2018).

To our knowledge, the present study is the first to investigate the oral microbiome of the defined population of patients having chronic HCV and individuals treated by DAAs. The only 4 studies (Aly et al., 2016; Heidrich et al., 2018; Inoue et al., 2018; Ponziani et al., 2018) that previously investigated patients having chronic HCV were solely concerned with gut microbiota. The studies concluded that gut microbiome of patients having chronic HCV have higher abundance of pathogenic bacteria. These bacteria are associated with higher levels of pro-inflammatory cytokines in the portal circulation, causing a "cytokine storm" and is believed to be initiated by bacterial translocation (Aly et al., 2016; Heidrich et al., 2018; Iebba et al., 2018; Inoue et al., 2018).

Likewise, a single study (Ponziani, Putignani, et al., 2018) investigated the gut microbiota of patients having chronic HCV before and after treatment with DAAs. Although unchanged after HCV clearance by old therapies (Interferon and Ribavirin) (Bajaj et al., 2016), after cure of HCV using DAAs, the gut microbiome was shifted in the direction approaching healthy samples through reduction in gut pathogenic bacteria (Ponziani, Putignani, et al., 2018). The present study adds to the aforementioned results (Ponziani, Putignani, et al., 2018), emphasizing that eradicating HCV by DAAs has favorable effects on the oral microbiome, similar to that in the gut microbiome.

The results of the present study revealed that the oral microbiome of buccal swabs of healthy individuals was highly occupied by *Streptococcus* species: *S. auralis*, *S. mitis*, *S. parasanguinus*, *S. infantis*, *S. oralis*, with low abundance of *Veillonella* species: *V. rogosea* and *V. denticariosi*. *Streptococcus* species are proven to interact with the epithelium, metabolize carbohydrates and prevent colonization of pathogenic bacteria, while *Veillonella* neutralizes the pH (Knapp et al., 2017). Therefore, this combination maintains a healthy condition. These results further explain the low values of diversity indices in the community of healthy controls.

On the contrary, in patients with chronic HCV, the community shows more variation with 17 unique species mainly of *Prevotella* and *Veillonella* genera, significant abundance of *Enterobacteriacea* and lower abundance of *Streptococcus* species. This explains the significantly higher alpha diversity compared to healthy controls.

Prevotella are gram-negative, obligate anaerobic rods that are involved in energy metabolism, carbohydrate, and lipid metabolism. They have pro-inflammatory actions by inactivating immunoglobulins, components of the complement system and antibacterial peptides (Kilian, 2018; Said et al., 2014). On the contrary, when found in abundance in the gut microbiome of patients having stage-4 HCV (Aly et al., 2016), *Prevotella* was further correlated to inflammatory response factors, namely IL-17 and T-helper 17.

As to the higher abundance of *Veillonella*, it was also reported in cases of primary sclerosing cholangitis, primary biliary cholangitis, and autoimmune hepatitis. At high abundance, it produces large amounts of lipopolysaccharides that stimulate inflammatory reactions. It was correlated to elevated levels of pro-inflammatory cytokines as IL-1β, IL-6, IL-8, and IL-12p70. In the same diseases, *Streptococcus* abundance was reduced, being correlated to the increase in levels of IL-1β and IL-8 (Abe et al., 2018).

The increase in abundance of *Enterobacteriaceae*—that normally resides in the lower gastrointestinal tract—was also detected in the saliva of patients having cirrhosis (Jasmohan S Bajaj et al., 2015). It may be attributed to deficient immune responses allowing for bacterial translocation of intestinal bacteria to reach the oral mucosa (Jasmohan S Bajaj et al., 2015).

The whole shift of the oral microbiome that was detected in HCV patients may be the indirect outcome of HCV particles infecting B-lymphocytes. The infected B cells secrete lower levels of IgA. Consequently, its modulatory action on microbiota is reduced, causing dysbiosis and abundance of pro-inflammatory bacterial species (Preveden et al., 2017).

Surprisingly, the results of the study further unveiled the similarity between the oral microbiome changes in patients having chronic HCV and that previously reported in patients having oral lichen planus (OLP) (Baek & Choi, 2017; Couvigny et al., 2015; He et al., 2017; Y. Li et al., 2020; Wang et al., 2016).

Chronic HCV is characterized by extra-hepatic manifestations including OLP, a chronic T-cell-mediated inflammatory oral disease. In OLP, T-lymphocytes attack basal keratinocytes of the oral mucosa. It was speculated that HCV-RNA resides in the oral epithelial cells, causing T cells to attack the keratinocytes, and inflammatory reactions proceed manifesting as OLP. However, no sufficient evidence supports this hypothesis (Baek & Choi, 2017).

An alternative explanation is proposed based on the results of the present, as well as, previous studies (Baek & Choi, 2017; Couvigny et al., 2015; He et al., 2017; Y. Li et al., 2020; Wang et al., 2016), proposing the oral microbiome shift as a causative factor for OLP.

Several studies investigated the oral microbiome of systemically healthy patients having OLP (Baek & Choi, 2017; Couvigny et al., 2015; He et al., 2017; Li et al., 2020; Wang et al., 2016). Interestingly, the results of these studies showed similar findings to the present study, revealing an increased microbial diversity and a significantly decreased *Streptococcus* genus abundance (Baek & Choi, 2017; Couvigny et al., 2015; He et al., 2017; Li et al., 2020; Wang et al., 2016). Besides, *S. salivarius* was proven to play an important role in affecting both inflammatory response and metabolic regulation of its host by regulating beneficial bacteria (Couvigny et al., 2015; Li et al., 2020). Therefore, its decreased abundance is highlighted as

a major player in the pathogenesis of OLP (Couvigny et al., 2015; Li et al., 2020).

The pattern of oral microbial shift in OLP was believed to cause intracellular infection of the basal keratinocytes with certain bacterial strains. These infected cells, thereby, become targets to T cells through liquifactive degeneration (Baek & Choi, 2017).

Linking the results of oral microbiome changes in cases of OLP to the—almost identical—findings in cases of chronic HCV of the present study, similar pathophysiology can be proposed for HCV-associated OLP. The new hypothesis proposes that dysbiotic bacteria (due to chronic HCV) infect the basal keratinocytes, thereby, inducing chronic T-cell reaction in the form of OLP.

Infection by HCV is proven to induce a systemic state of inflammation that affects all mucosal barriers causing "global mucosal-immune change" (Ling et al., 2015). A vicious circle, then proceeds where systemic inflammation induces more pronounced oral dysbiosis, and, in return, dysbiosis causes more aggravated systemic inflammation, a cycle that maintains a chronic behavior to both diseases: chronic HCV and HCV-associated OLP.

Lastly, with the clearance of HCV by DAAs, the consequences of the infection are expected to be eradicated as well. However, the results of the present study show that the extreme shift detected in HCV-positive patients was not fully restored after DAA treatment. As HCV infection was linked to the increase in the abundance of pathogenic species at the expense of *Streptococcus* species, treatment by DAAs was linked to an increase in *Streptococcus* species and decrease in *Prevotella* species, but not reaching the healthy levels. The alpha diversity indices were still elevated compared to healthy controls, but rather insignificantly, and the bacterial community showed partial similarity and shared members with both that of HCV-positive and healthy individuals' microbiome.

These results can be viewed in the light of previous evidence (Ponziani, Putignani, et al., 2018) that DAAs succeed in modulating the gut microbiota, though failing at reversing the intestinal permeability. By doing so, the gut microbiome was shifted in the direction of the normal balance, but not reaching it owing to the continuous state of bacterial dislocation. Similar explanation might be applicable to the action of DAAs on oral microbiome.

The present study clarified the extent of the shift in oral microbiome in cases of chronic HCV and its partial restoration after DAA therapy. By doing so, the inflammatory processes induced by the dysbiotic community are reduced, shifting to a healthier oral–gut–liver axis. However, an inherent limitation in the study design is its ability to prove only association between chronic HCV with dysbiosis, and DAA treatment with intermediate state of microbiome, rather than causation. Future longitudinal studies may reveal the map of oral microbiome change at different time points following treatment. Moreover, the scope of the study was only to investigate the oral microbiome; we did not examine or correlate the gut microbiome of the included participants. Lastly, the sample size may be considered limited. However, the present work should be noted as a preliminary study for a novel point of research.

In conclusion, compared to healthy individuals, patients having chronic HCV have a significantly dysbiotic oral microbiome which shifts toward abundance of bacteria that activate pro-inflammatory pathways, inactivate anti-inflammatory pathways, and reflect increased intestinal permeability and bacterial translocation. With HCV clearance by DAAs, the oral microbiome is partially restored into a community that is midway between the diseased and healthy ones.

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PATIENT CONSENT STATEMENT

Informed consent was obtained from all participants in the study.

CLINICAL TRIAL REGISTRATION

The study protocol was registered on ClinicalTrials.gov with an ID of NCT03682549.

CONFLICT OF INTEREST

None to declare.

AUTHOR CONTRIBUTIONS

Ayat Gamal-AbdelNaser: Conceptualization; Methodology; Writing-original draft. **Waleed S Mohammed:** Data curation; Software; Visualization; Writing-review & editing. **Mahmoud ElHefnawi:** Data curation; Software; Writing-review & editing. **Mohamed AbdAllah:** Conceptualization; Investigation; Methodology; Supervision; Writing-original draft. **Aisha Elsharkawy:** Conceptualization; Investigation; Methodology; Supervision; Writing-review & editing. **Fat'heya M Zahran:** Conceptualization; Supervision; Writing-review & editing.

PEER REVIEW

The peer review history for this article is available at https://publons.com/publon/10.1111/odi.14007.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in NCBI's sequence reads archive at https://www.ncbi.nlm.nih.gov/bioproject/PRJNA725877, reference number PRJNA725877.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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